heat_loss_watt = 332.86
solar_gain_watt = 1783.15
design-day-load (thermal): known
loss_btu_per_hour = 2076.43
solar_gain_btu_per_hour = 6084.37
heating-season (thermal): known
season_loss_btu = 9834109.4
season_solar_btu = 6159297.14
solar_share_percent = 62.63
fuel_units_needed = 0.26
joist-deflection (structural): known
deflection_mm = 4.26
span_over_deflection = 571.7
bending_stress_mpa = 4.65
gravity-load-path (structural): unknown
the ground reaction is unknown: post@3048,2438,-1219~140x140: roof@0,0,3658:3~3658x5486 may rest on it: it is in none of the roles panel_pieces, lumber_pieces, occupied_floors, ground and openings, so the load path is given no size or material for it; 0 of 25 pieces carry a known force
pieces = 25
pieces_weight_known = 9
pieces_force_known = 0
sources: sip.prs610_panel_thickness (published); sip.prs610_core_thickness (published); osb.gp_7_16 (published); sip.sipa_definition (published); wt.syn_osb_sheet (inferred); e1295.core (published); lum.nds.fv_spf_no2 (visual-read); lum.nds.cf_2x6 (visual-read); lum.nds.cr (published)
header-check (structural): known
reference design values with no NDS adjustment factor; the roof carries the ground snow load unreduced; a header's load is the roof over its wall and the joists landing at its head, not its own weight nor the wall or glass above it; joists land part way up door@610,4819, glazing@610,2495,1219, with no header under them
headers = 3
headers_over_one = 0
utilisation_max = 0.61
sources: lvl.width (published); lvl.depth (published); gl.prop.lvl_2_0e_2600 (visual-read); gl.prop.lvl_density_unpublished (unpublished)
work-at-published-rates (tasks): known
post_holes_hours = 2
floor_hours = 1
walls_hours = 5.72
roof_hours = 1.91
measured_hours = 10.63
panel_usd = 6293.36
pieces_not_priced = 4
not_measured = set-posts (task.t2.hours_set_post_unpublished); lay-subfloor (task.t4.hours_subfloor_unpublished); cut-openings (task.t6.hours_field_opening_unpublished); set-windows-and-door (task.t7.hours_windows_door_unpublished); frame-loft (task.t9.hours_loft_measured_unpublished); install-boiler (task.t11.hours_boiler_hydronic_total_manhours_unpublished)
sources: task.t1.hours_hand_ramirent (published); task.t5.hours_premier_cs1_sqft (inferred); task.t5.hours_premier_cs1_sqft (published); task.t8.hours_premier_cs3_roof_rate (published); sp.acme_other_thicknesses (published); sp.acme_1025_price (published)

RULES, EACH WITH ITS STATUS

posts-under-floor [foundation]: PASS: every floor corner has a post under it; source: integrity: a floor on posts needs a post at each corner
walls-enclose-floor [walls]: PASS: walls close every edge of the floor; source: integrity: a shell closes every edge of its floor
roof-is-over-floor [roof]: PASS: a roof stands over the whole floor; source: integrity: a roof stands over the floor it keeps dry
egress-door-size [doors]: PASS: the egress door is at least as large as the limit; source: Michigan Residential Code 2015 R311.2, research row code.r311_2_door: side-hinged, 3 ft by 6 ft 8 in
posts-below-frost-line [foundation]: PASS: the posts are buried below the frost line; source: assumed: the frost depth the site gives; a footing must sit below it
roof-covers-floor [roof]: PASS: the roof covers the floor; source: Jim's what: the roof keeps the floor dry
headers-within-capacity [header_check]: PASS: every header is within its capacity; source: the header's catalog design values: bending, shear and its deflection limit
joists-span-the-deck [joists]: PASS: the joists span the deck at the stated spacing; source: integrity: a deck needs joists from edge to edge at the stated spacing

GRAVITY LOAD PATH, PER POST

status: unknown; the ground reaction is unknown: post@3048,2438,-1219~140x140: roof@0,0,3658:3~3658x5486 may rest on it: it is in none of the roles panel_pieces, lumber_pieces, occupied_floors, ground and openings, so the load path is given no size or material for it; 0 of 25 pieces carry a known force
post@3048,2438,-1219~140x140: ground reaction unknown (roof@0,0,3658:3~3658x5486 may rest on it: it is in none of the roles panel_pieces, lumber_pieces, occupied_floors, ground and openings, so the load path is given no size or material for it)
post@3048,4876,-1219~140x140: ground reaction unknown (roof@0,0,3658:3~3658x5486 may rest on it: it is in none of the roles panel_pieces, lumber_pieces, occupied_floors, ground and openings, so the load path is given no size or material for it)
post@610,2438,-1219~140x140: ground reaction unknown (roof@0,0,3658:3~3658x5486 may rest on it: it is in none of the roles panel_pieces, lumber_pieces, occupied_floors, ground and openings, so the load path is given no size or material for it)
post@610,4876,-1219~140x140: ground reaction unknown (roof@0,0,3658:3~3658x5486 may rest on it: it is in none of the roles panel_pieces, lumber_pieces, occupied_floors, ground and openings, so the load path is given no size or material for it)

HEADER UTILISATIONS

status: known; largest 0.61
door@610,4819: utilisation 0.04; bending 0.29 MPa, shear 0.07 MPa, deflection 0.02 mm; load 2.1 N/mm against capacity 57.63 N/mm
glazing@610,2495,1219: utilisation 0; bending 0 MPa, shear 0 MPa, deflection 0 mm; load 0 N/mm against capacity 28.82 N/mm
glazing@610,2495,2438: utilisation 0.61; bending 9.7 MPa, shear 1.2 MPa, deflection 2.89 mm; load 17.6 N/mm against capacity 28.82 N/mm

BUILD SIMULATION TOTALS

ended quiescent; clock step 1 minute until 1440
finish_time = 22
labour_hours = 14
learning = 0.6
idle = 122
score = -119.4
final stocks:
budget_usd = 13706.64
cost_usd = 6293.36
floor_panel_ft2_set = 63.98
pieces_not_priced = 4
post_holes_dug = 4
roof_panel_ft2_set = 216.01
wall_panel_ft2_set = 366.02
flows:
dig-post-holes: ran 4 times; moved pieces_not_priced 4, post_holes_dug 4
set-floor-panels: ran 2 times; moved budget_usd -584.77, cost_usd 584.77, floor_panel_ft2_set 63.98
set-wall-panels: ran 12 times; moved budget_usd -3345.46, cost_usd 3345.46, wall_panel_ft2_set 366.02
set-roof-panels: ran 4 times; moved budget_usd -2363.13, cost_usd 2363.13, roof_panel_ft2_set 216.01
set-posts: ran 0 times; moved
lay-subfloor: ran 0 times; moved
cut-openings: ran 0 times; moved
set-windows-and-door: ran 0 times; moved
frame-loft: ran 0 times; moved
install-boiler: ran 0 times; moved

ASSUMPTIONS, EVERY SETTING WHOSE REASON SAYS IT IS ASSUMED

floor.thickness_mm = 114 mm; reason: assumed: a 4-1/2 in (114 mm) SIP floor panel, the wall standard's thickness, research row sip.prs610_panel_thickness; Jim's floor build-up gives it
weather.u_value = 2.8 w_per_m2_k; reason: assumed: typical double glazing, not chosen by Jim
weather.solar_heat_gain_coefficient = 0.6 ratio; reason: assumed: typical double glazing, not chosen by Jim
weather.inside_celsius = 20 celsius; reason: assumed design day
weather.outside_celsius = 0 celsius; reason: assumed design day
weather.sun_watt_per_square_metre = 500 w_per_m2; reason: assumed clear-sky peak; the solver applies it to every side

PROJECT	SHEET	SHEET NUMBER	DATE
Small house on posts, from Jim's pencil sketch	**Design summary**	S-1 (page 2 of 3)	2026-10-20
CONFIG COMMIT	ENGINE PIN	SCALE	SOURCE
c2a9df0	we 02d17b7; rendered with we 073b001	not to scale	world-engine geometry; no line is drawn by hand